Supplementary Information
Road-structured constitutive coupling between source–sink patterns and anisotropic urban routing

Contents

Supplementary Methods **2**

Note S1 Data versions, observation regimes and frozen splits . . . 2
Note S2 Fields, road registration, estimation and uncertainty . . . 3
Note S3 Joint spectra and coherent response . . . 5
Note S4 Anisotropy and matched-capacity media . . . 9
Note S5 Baseline and comparison protocols . . . 11

Supplementary Results **11**

Note S6 Held-out coupling and baseline results . . . 12
Note S7 Joint-spectrum, daily-score and coherent-response results . . . 14
Note S8 Three-city anisotropy results . . . 15
Note S9 Road specificity and matched-capacity structural results . . . 15

Supplementary Reproducibility **16**

Note S10 Artifact map and reproducibility . . . 16

Supplementary Methods

Notes S1–S5 specify the three-city data, fields, road registration, estimators, scores, uncertainty, joint spectra, coherent response, anisotropy, matched-capacity media and comparison protocols. Figures are numbered by first appearance and interpreted in the corresponding results Notes.

Note S1 Data versions, observation regimes and frozen splits

The focal records are Porto taxi trajectories from the ECML/PKDD 2015 challenge [1], San Francisco Cabspotting trajectories (CRAWDAD `epfl/mobility`, version 2009-02-24) [2], and an archived 2014 DiDi ride-hailing sample from Chengdu [3], with regularly spaced 30-second fixes in the aligned protocol. Table S1 separates releases, observation regimes and frozen date protocols. Errors are never pooled across cities or observation regimes.

Table S1: Three-city observation regimes and frozen date splits. TRAIN/VAL/TEST counts are dates. San Francisco has five blocked folds, each 9/3/3. Chengdu's three protocols are related but not interchangeable.

Object	Data version or release	Regime and frozen protocol	Split	Lattice	Hours
Porto	ECML/PKDD 2015	Taxi; 252 eligible dates; chronological	151/50/51	32×32	00–23
San Francisco	Cabspotting v. 2009-02-24	Taxi; 15 dates; five blocked folds	9/3/3 per fold	32×88	00–23
Chengdu: P/D, spectra, anisotropy	Archived DiDi 2014 sample	Continuous-observation subset; 14 dates	8/3/3	60×60	06–23
Chengdu: structural null	Same aligned grid	All 19 valid dates; full aligned observation pool, not a P/D fit	–	60×60	aligned pool
Chengdu: matched capacity	Same archived corpus	Chronological 18-date protocol; four TEST dates 26–29 Aug. 2014 use alternate-hour delivery, nine recorded hours/date	10/4/4	60×60	alternating subset of 06–23

Chengdu's 14-date subset has three TEST dates in the continuous-observation regime and supplies the P/D values in Table S2. Its 18-date chronological split instead supplies the matched-capacity experiment: the four held-out dates fall wholly in the alternate-hour delivery phase. The 19-date pool is used only to compare representation subspaces. These different date counts and observation regimes must not be merged. The design cannot separate the influence of date count, delivery phase and channel on the matched-capacity effect; its four-date interval is only a descriptive small-sample summary.

The nominal cell size is 250 m on the focal aligned grids. Successive trajectory segments are walked through aligned cells and accumulated as signed interior-face crossings in 60-minute windows. No map matching is applied. Boundary faces are excluded and the archived activity mask is fixed. The archive does not establish that this mask was derived from TRAIN alone: its TEST

provenance is therefore unknown, and the registered support is treated as a fixed transductive spatial convention rather than demonstrably TEST-independent preprocessing. The flux-derived source is not an independently measured trip-production field. Unrecorded raw-data schema or preprocessing fields are not reconstructed from their names or presumed to be available.

Attribution, licensing and release checks. Road geometry derives from the drivable OpenStreetMap layer [4]; maps and derived road products retain the attribution © *OpenStreetMap contributors*. The current archive does not provide sufficient per-extract metadata to verify the OSM snapshot dates/versions, the licence notices accompanying every derived artifact, or all downstream redistribution obligations. Those items must be checked against the source extracts before public release; no city-specific version or licence clearance is invented here. The exact public release/version and redistribution terms of the archived Chengdu sample likewise remain unverified. Public data availability does not imply that individual-level trajectories are included in the export.

Note S2 Fields, road registration, estimation and uncertainty

Discrete fields and conservation. Let $F_t \in \mathbb{R}^{n_f}$ be the observed signed crossing field on interior faces; a prediction is distinguished by a hat. The cell–face incidence matrix $D \in \mathbb{R}^{n_c \times n_f}$ maps face flux to net cell outflow under the frozen orientation. Thus

$$x_t = DF_t, \qquad P_0 = I - D^\dagger D, \qquad F_t^\circ = P_0 F_t, \qquad DF_t^\circ = 0. \tag{S1}$$

The observed source–sink balance and divergence-free routing field give the spatially orthogonal decomposition $F_t = D^\dagger x_t + F_t^\circ$. Orthogonality at an instant does not imply statistical independence across observations. The minimum-norm lift uses a gauge-pinned Laplacian solve rather than an explicitly formed Moore–Penrose inverse; on the zero-sum compatible source space the two constructions coincide. Conservation is imposed by construction or projection, not learned.

Road registration. The registered road graph has active cells as nodes and registered road segments as edges. If B_R is its node–edge incidence, A_R lifts signed road-edge flows to interior faces, and Π aggregates cells to nodes, registration satisfies $DA_R = \Pi B_R$ in integer arithmetic. For a cycle basis Z_R with $B_R Z_R = 0$,

$$U_R = \mathrm{orth}(A_R Z_R), \qquad U_R^\top U_R = I, \qquad DU_R = 0. \tag{S2}$$

The signed entries of A_R count open-segment crossings in the frozen face orientation. Geometry, domain, mask and registration determine the basis; mobility values, dates and folds do not. This independence is conditional on the registered spatial conventions. The observed road-coordinate state is $y_t = U_R^\top F_t^\circ$. The physical lift $C_R = U_R K_R P_x$, with $P_x = DD^\dagger$, is used for the local response tensors in Note S4.

Response estimation. For any source or target v_t, TRAIN phase means and departures are

$$\bar{v}_\phi = \frac{1}{n_{\mathrm{tr},\phi}} \sum_{t \in \mathrm{TRAIN}:\phi(t)=\phi} v_t, \qquad \delta v_t = v_t - \bar{v}_{\phi(t)}. \tag{S3}$$

Stack $\delta x_t^\top$ and $\delta y_t^\top$ as rows of X and Z. The road-coordinate ridge response is

$$K_R^\star = \arg\min_K \{\|Z - XK^\top\|_F^2 + \lambda \|K\|_F^2\} = Z^\top X (X^\top X + \lambda I)^{-1}, \tag{S4}$$

with $\lambda = \kappa s_{\max}(XX^\top)$ and $\kappa \in \{10^{-10}, 10^{-8}, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$. Validation selects κ in the scoring frame of each comparison arm. The structural-characterization estimators use $\kappa = 0.01$ in Porto, 0.1 in San Francisco, 0.1 in Chengdu's D frame, and 0.001 for Chengdu's P-frame sensitivity. Each city is fitted separately; these settings are not a cross-city transfer rule.

P, D and CLIM scores. Let $T_t = U_R^\top F_t^\circ$ on ROAD support and $T_t = F_t^\circ$ on FACE support. ROAD scoring projects full-face predictions into road coordinates; FACE scoring lifts road predictions to the complete routing field, so their denominators differ. Both fitted maps use TRAIN departures. P selects on raw VAL inputs and targets and predicts $B_{\mathrm{P}} x_t$ without a fitted intercept; D centres VAL/TEST with TRAIN phase means and predicts $B_{\mathrm{D}} \delta x_t$:

$$
\begin{aligned}
R^2_{\mathrm{P}} &= 1 - \frac{\sum_{t \in \mathrm{TEST}} \|T_t - B_{\mathrm{P}} x_t\|^2}{\sum_{t \in \mathrm{TEST}} \|T_t\|^2}, \\
R^2_{\mathrm{D}} &= 1 - \frac{\sum_{t \in \mathrm{TEST}} \|\delta T_t - B_{\mathrm{D}} \delta x_t\|^2}{\sum_{t \in \mathrm{TEST}} \|\delta T_t\|^2}.
\end{aligned}
\tag{S5}
$$

Numerators and denominators are pooled over TEST observations and folds within each city. The denominator is target energy, not variance about a TEST mean. CLIM is the source-free prediction $\bar{T}_{\phi(t)}$ on the same support and samples. For the P-frame physical display in Figure S6, the observed field is $U_R T_t$ and the prediction is $U_R B_{\mathrm{P}} x_t$, using raw TEST inputs and targets without adding a phase climatology or substituting a D departure.

Full-face ridge fits F_t° and projects its predictions with P_0. POD regression estimates its output basis from TRAIN departures only and selects rank and regularization on VAL in the common FACE metric. No TEST observation selects a basis, rank or regularization. These linear controls and the matched-capacity protocol in Note S4 are distinct experiments.

Nulls and uncertainty. The rank-matched structural null fixes the ambient space and basis rank but not the subspace. Let K_D be an orthonormal basis of $\ker D$. Each draw uses $G_{ij} \overset{\mathrm{iid}}{\sim} N(0,1)$, $Q = \mathrm{orth}(G)$ and $U_b = K_D Q$ at $\mathrm{rank}(U_R)$. The statistic is

$$
E(U) = 1 - \frac{\sum_t \|U^\top F_t^\circ\|^2}{\sum_t \|F_t^\circ\|^2}, \qquad \Delta E_b = E(U_b) - E(U_R),
\tag{S6}
$$

evaluated on each city's full aligned observation pool without a source-response fit. It tests representation specificity, not source-based prediction.

The primary three-city D scores in Table S2 have no reported paired-date interval or pairing-null result. Per-date displays do not supply that missing uncertainty quantification. The neural paired-date contrast uses 9,999 bootstrap resamples, seed 20261003, dates resampled within folds and Holm adjustment within city/support. The matched-capacity implementation uses 10,000 multinomial paired-date resamples with seed 23 and summarizes the date-mean contrast. Faces are not resampled. Chengdu's matched-capacity interval resamples only four alternate-hour TEST dates and is a small-sample descriptive interval.

Note S3 Joint spectra and coherent response

Joint temporal–spatial spectrum. For raw field $v_{d,j}$ on a TRAIN-date block of length N, a unitary one-sided discrete Fourier transform gives

$$\begin{aligned} Z_{d,m} &= N^{-1/2} \sum_{j=0}^{N-1} v_{d,j} e^{-2\pi i m j/N}, \qquad f_m = m/N, \\ P_m &= w_m \langle \|Z_{d,m}\|^2 \rangle_d, \qquad J_{m,k} = \frac{P_m}{\sum_l P_l} \frac{s_{m,k}^2}{\sum_a s_{m,a}^2}. \end{aligned} \tag{S7}$$

Here $s_{m,k}$ are singular values of the date-by-space coefficient matrix, $w_0 = w_{N/2} = 1$, and interior weights are two. The transform retains $m = 0$, so the spectra include each date's phase mean. Porto and San Francisco use $N = 24$; Chengdu uses $N = 18$ for 06–23 h without night imputation or zero filling. San Francisco averages P_m and within-frequency mode shares separately across TRAIN folds before multiplication. Normalization uses the full weighted spectrum before the eight-mode display cut in Figure S1. Spatial modes are frequency-specific singular modes, not geographical Fourier wavelengths.

Chengdu has ten single-sided bins $m = 0, \ldots, 9$ at $m/18$ cycles per observed hour. The periodic extension joins the ends of the observed block; it does not infer the unobserved night. A 24-hour sinusoid is off this grid, so the first nonzero bin is not identified as the once-daily harmonic. Zero padding to 24 samples would change the frequency grid and redistribute shares, not establish zero night-time flow.

Coherent TRAIN/TEST response. The frozen P-selected operator is applied without refitting a frequency response. Each date is transformed with the same unitary DFT support as the descriptive spectrum: 24 points in Porto and San Francisco and 18 points in Chengdu. Let $a_m^{(\mathcal{S})}$ and $b_m^{(\mathcal{S})}$ be observed and predicted routing coefficients averaged over dates in split $\mathcal{S} \in \{\text{TRAIN}, \text{TEST}\}$. For each split and single-sided bin m,

$$\gamma_m^{(\mathcal{S})} = \frac{|a_m^{(\mathcal{S})*} b_m^{(\mathcal{S})}|}{\|a_m^{(\mathcal{S})}\| \|b_m^{(\mathcal{S})}\|}, \qquad \rho_m^{(\mathcal{S})} = \frac{\|b_m^{(\mathcal{S})}\|}{\|a_m^{(\mathcal{S})}\|}. \tag{S8}$$

The absolute inner product makes γ_m insensitive to a common phase rotation. TRAIN values describe the fit sample; TEST values are held-out. San Francisco summaries are fold means and ranges. These coherent averages complement, rather than replace, observation-level P/D scores. Chengdu evaluates all ten bins $m = 0, \ldots, 9$, exactly as in its joint spectrum. Figure S3 displays both splits without treating the TRAIN comparison as independent validation.

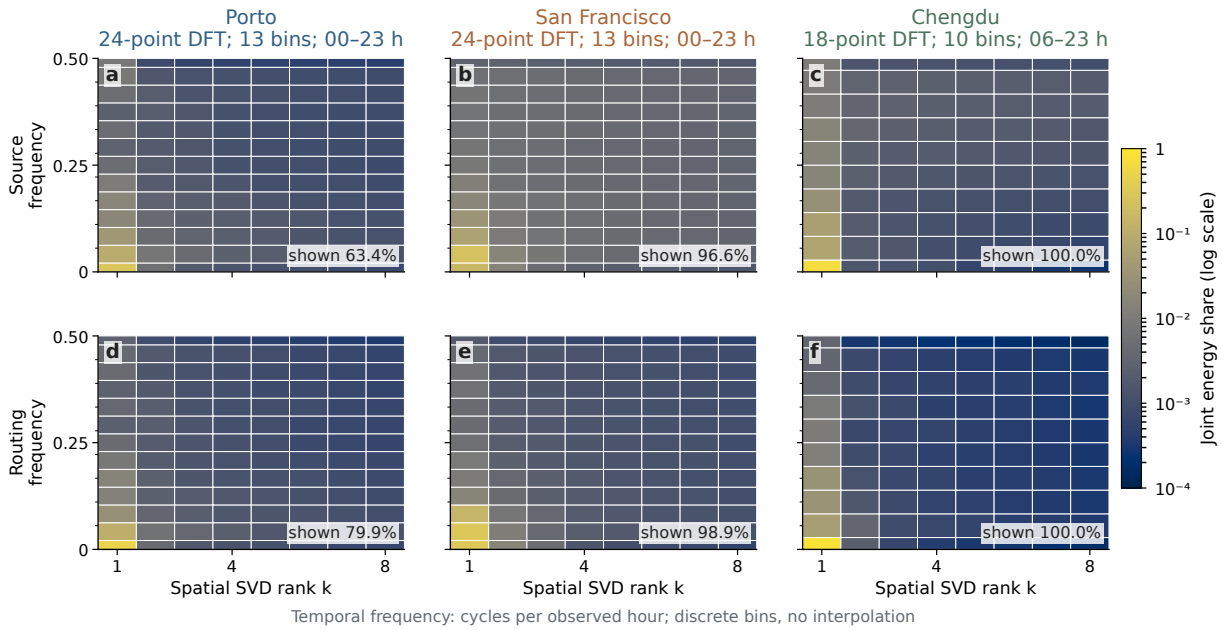


Figure S1: Three-city two-dimensional joint spectra. Joint energy share over temporal frequency and frequency-specific spatial mode rank for source and full-face routing fields in Porto, San Francisco and Chengdu. The eight-mode display cut is applied after full-spectrum normalization. Porto and San Francisco use 24-point DFTs; Chengdu uses an 18-point observed-block DFT with no night imputation. San Francisco averages frequency energies and within-frequency mode shares separately across TRAIN folds. Spatial rank is not geographical wavelength.

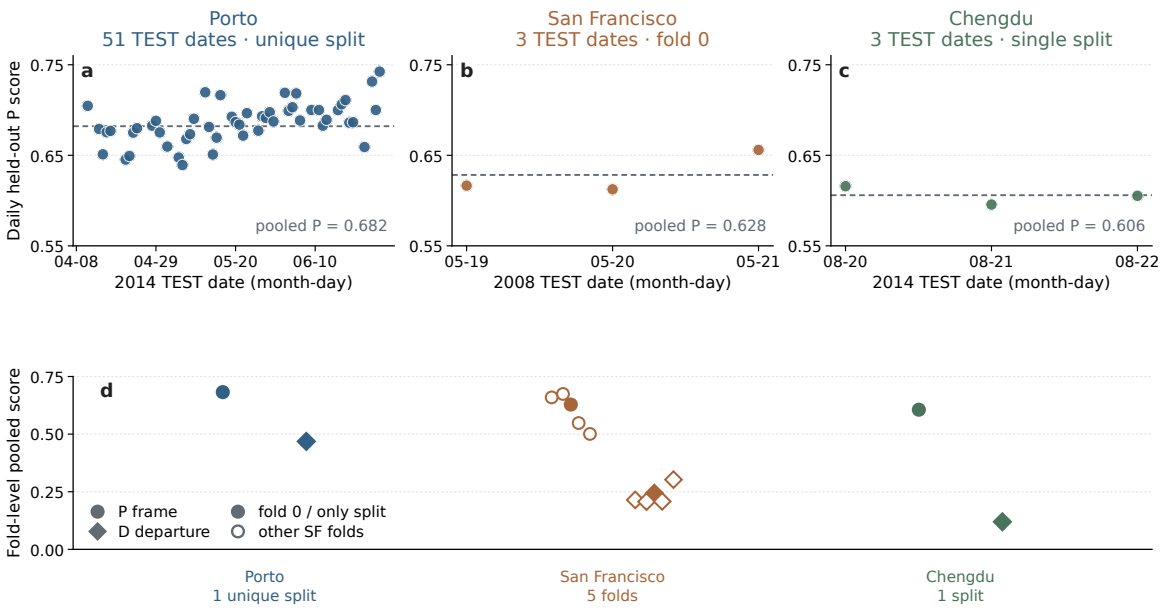


Figure S2: Three-city held-out daily P/D scores. Date-specific P and D values for Porto, San Francisco and Chengdu under the frozen splits in Table S1. San Francisco retains its five-fold structure; Chengdu has three TEST dates in the continuous-observation protocol. Daily values complement the pooled scores in Table S2; their spread is not a paired-date confidence interval.

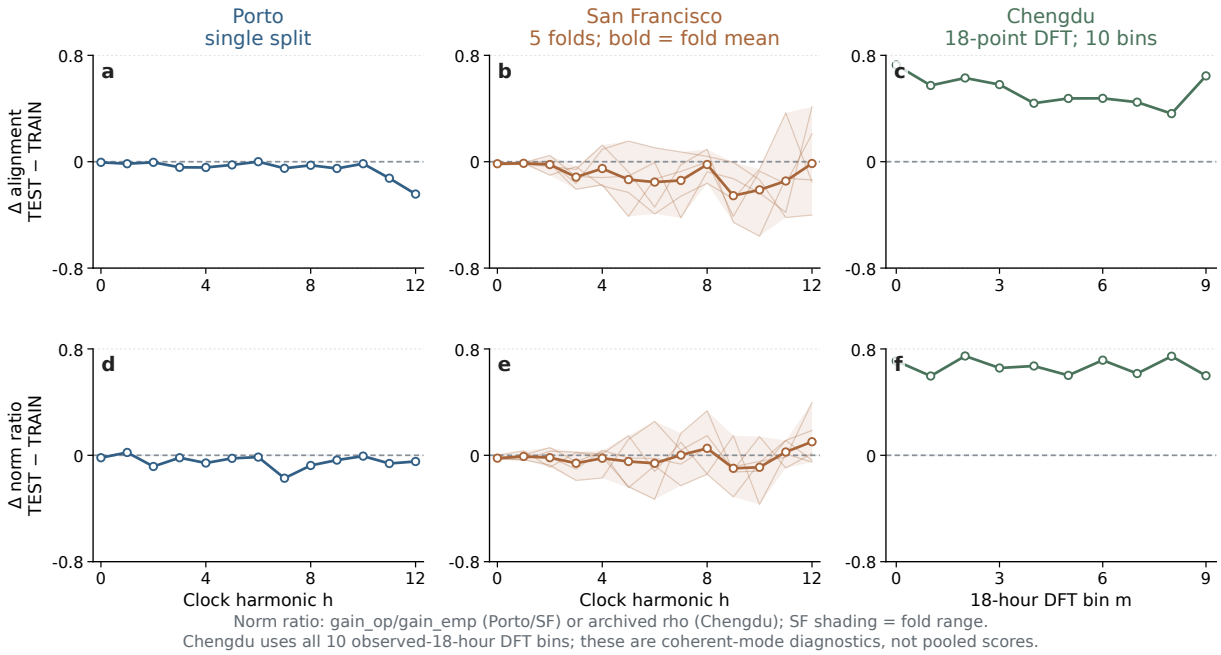


Figure S3: Three-city coherent response on TRAIN and TEST. Direction alignment γ_m and predicted-to-observed norm ratio ρ_m for Porto, San Francisco and Chengdu under the frozen P-selected operator. TRAIN values describe the fit sample; TEST values are held-out. San Francisco shows fold means and ranges. Each date uses the same unitary DFT support as the descriptive spectrum; Chengdu evaluates all ten single-sided bins of its 18-point transform. Alignment is insensitive to a common phase rotation.

Note S4 Anisotropy and matched-capacity media

Local tensors and axial orientation. For each regular interior cell, L_c maps signed face flux to two physical components. Define

$$\begin{aligned}\Sigma_C(c) &= L_c C_R C_R^\top L_c^\top,\\ \Sigma_U(c) &= L_c U_R U_R^\top L_c^\top,\\ A(c) &= \begin{cases}(\lambda_1(c)-\lambda_2(c))/(\lambda_1(c)+\lambda_2(c)), & \lambda_1(c)+\lambda_2(c)>0,\\ 0, & \lambda_1(c)+\lambda_2(c)=0.\end{cases}\end{aligned} \tag{S9}$$

Here $\lambda_1 \geq \lambda_2$ are eigenvalues of the relevant tensor. The principal eigenvector defines an axis θ modulo π, not a directed arrow. San Francisco uses per-cell mean anisotropy and a nematic mean of $e^{2i\theta}$ over five folds. Directional coherence is

$$S_{\text{nem}} = \frac{\left|\sum_c w_c A_C(c) e^{2i\theta_C(c)}\right|}{\sum_c w_c A_C(c)}, \tag{S10}$$

with $w_c = 1$ or $w_c = \operatorname{tr}\Sigma_C(c)$.

Structural-axis agreement is the joint-anisotropy estimand

$$G_\theta = \frac{\sum_c A_C(c)A_U(c)\cos[2(\theta_C(c)-\theta_U(c))]}{\sum_c A_C(c)A_U(c)}, \qquad A_C(c)A_U(c)>0, \tag{S11}$$

on regular interior cells for which both anisotropies are nonzero. On the regular interior grids $L_c L_c^\top = 2I$, so metric whitening is a no-op and grid anisotropy vanishes there; anisotropic boundary-cell metrics are excluded. An orthogonal road-coordinate rotation leaves the physical tensor unchanged when basis and coefficients are transformed together. Under a rigid physical rotation Q,

$$\Sigma_C(c) \mapsto Q\Sigma_C(c)Q^\top, \qquad v_C(c) \mapsto Qv_C(c), \tag{S12}$$

while eigenvalues and scalar anisotropy are invariant. The axial vector is covariant, not numerically unchanged in fixed physical coordinates.

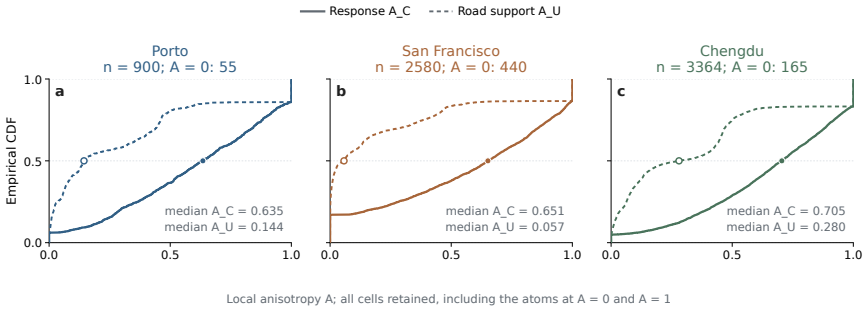


Figure S4: Three-city local anisotropy distributions. Empirical cumulative distribution of $A = (\lambda_1 - \lambda_2)/(\lambda_1 + \lambda_2)$, with $A = 0$ assigned to a zero tensor, over regular interior cells for road support A_U (dashed) and fitted response A_C (solid); circles mark medians. Both curves use the same cells and per-cell definition. Endpoint steps are real atoms: cells without directional energy enter at $A = 0$ (5–17%, identical cell sets for A_U and A_C), and one-dimensional tensors enter at $A = 1$ (6–16%). They are not normalization artefacts.

Matched-capacity media. The compared media are a road medium from an external GIS raster, a mobility-inferred medium estimated from TRAIN mobility observations, and a permuted mobility medium. At capacity R, each per-face source pattern is averaged over R spatial blocks and passed through the verified mean-1/floor normalization

$$w_{S,f}^{(R)} = \max\left(\frac{\overline{W}_{S,f}^{(R)}}{\langle \overline{W}_{S}^{(R)} \rangle_{\text{interior faces}}}, 0.02\right), \tag{S13}$$

before the archived minimum-energy field lift. Thus R is the nominal number of block values before normalization, not the effective degrees of freedom: mean-1 normalization removes at least one independent degree, and the floor can reduce it further. With $R = 1$ every medium is constant, and $R = n_{\text{int}}$ assigns one pre-normalization value per interior face. Permutation preserves the mobility medium's values, partition and nominal block count while permuting its spatial arrangement. Road and mobility media do not have identical information inputs; equal R matches capacity, not information.

The score is computed per held-out date against that date's own target mean and then averaged with equal date weight:

$$R^2_{\text{cond},d} = 1 - \frac{\sum_{t\in d}\|T_t - \widehat{T}_t\|^2}{\sum_{t\in d}\|T_t - \langle T\rangle_d\|^2}. \tag{S14}$$

It is an archived date-centred statistic on the raw routing target, not on D's TRAIN-phase departure target, and is distinct from P and D in Eq. S5. Figure S5 reports gains over the constant $R = 1$ medium.

The nested grids are Porto $R = (1, 4, 16, 64, 256, 1984)$ with block sides $(0, 16, 8, 4, 2, 1)$; San Francisco $R = (1, 3, 12, 44, 176, 704, 5512)$ with block sides $(0, 32, 16, 8, 4, 2, 1)$; and Chengdu $R = (1, 4, 16, 64, 225, 900, 7080)$ with block sides $(0, 32, 16, 8, 4, 2, 1)$. Side zero denotes the single constant block. The final identity partition gives $R = n_{\text{int}} = (1984, 5512, 7080)$. San Francisco uses its occupancy-defined channel and per-fold basis; Chengdu has only the archived all-motion channel and uses the frozen per-fold-basis rule on its chronological 10/4/4 split.

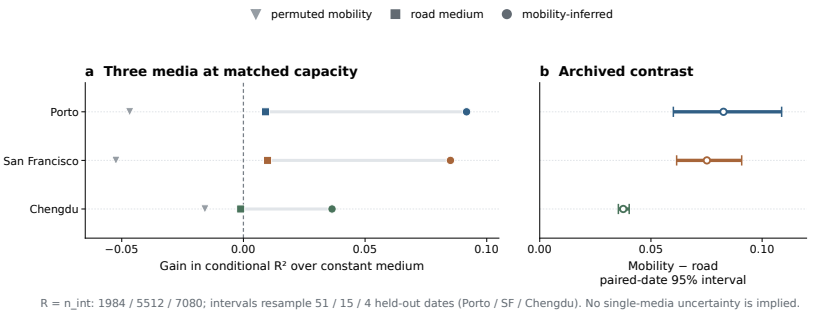


Figure S5: Three-city media at matched capacity. Gain in held-out conditional R^2 over the constant $R = 1$ medium for the permuted mobility medium, road medium and mobility-inferred medium at $R = n_{\text{int}}$ (1984, 5512 and 7080 nominal pre-normalization block values). The score is the equal-date mean of within-date conditional R^2, distinct from P and D. Constant-medium values are 0.613, 0.743 and −0.030. Permutation preserves values, partition and nominal block count. Chengdu resamples four alternate-hour dates on a single all-motion channel; its points are small-sample summaries.

The definitions above are limited to code and frozen outputs available in the export. The exact construction of every named source pattern and the full field-building implementation are not self-

contained; the missing modules, legacy root and unavailable inputs are itemized in Note S10. The archived passing-cell record is not a runnable reproduction gate in the present export.

Note S5 Baseline and comparison protocols

Graph WaveNet. The completed adaptation [5] estimates the contemporaneous source-to-routing map on the frozen date splits. The implemented `cc_lib.anomaly` mask, rather than its misleading "every row" docstring, defines a P/FACE score: TRAIN inputs and targets are phase-centred, whereas VAL/TEST inputs and targets remain raw. Predictions are projected with P_0, so zero divergence is imposed rather than learned.

The archived loop constructs the network before seeding the training routine, leaving weight initialization uncontrolled. The sealed protocol declares one integer before construction, seeds `random`, NumPy and PyTorch, and reruns the full four-configuration grid with VAL selection repeated inside every seed. Porto and Chengdu use five seeds; San Francisco uses four seeds common to all five folds. The archived cells remain identifiable as single draws. The final duplicate Porto fold-0 per-date record is authoritative. Graphormer's separate 60-epoch reduced-budget run is excluded from the matched comparison and does not bound its model class.

Gravity and OD overlap. The doubly constrained gravity model estimates OD flows from TRAIN margins and distance, selecting friction on VAL. Its native common-part-of-commuters score at 4×4-cell aggregation is an OD-overlap measure, not a routing R^2, and the two are not numerically comparable. Radiation and Deep Gravity were not run because the required population/opportunity or covariate inputs were not supplied; no numerical result is attributed to them [6, 7].

All-or-nothing loading. AON uses time-shortest paths with archived road length/speed costs, a 150 m endpoint-snap threshold, and no congestion equilibrium or capacity calibration. Gravity loading retains the top 150,000 zone pairs under the archived combined-demand rule. A TEST-fitted global amplitude is diagnostic rather than deployable; the fixed-amplitude arm uses $\alpha = 1$. Exact-TEST-OD loading has additional demand information and is not a source-only prediction. AON does not stand in for Wardrop user equilibrium [8], which was not run.

Chengdu OD is cut from the corpus occupancy flag as maximal occupied runs, with frozen cleaning breaks (fix gap > 15 min, implied speed $> 120\,\mathrm{km\,h^{-1}}$, displacement > 3000 m) and retention above 60 s and 100 m. It is not inferred from a displacement heuristic. The observed field counts crossings during $[t_k, t_k + 3600\,\mathrm{s})$, whereas the loaded field routes every trip whose origin lies in window k. Moreover, Chengdu's observed field is marched from every fix while its OD uses occupied runs only; these overlapping but different activity sets are part of the comparison protocol.

Supplementary Results

Notes S6–S9 report held-out coupling, daily scores, coherent response, anisotropy and structural tests. The three-city P-frame physical reconstruction is a qualitative complement to the complete held-out evaluation.

Table S2: Three-city P/D/CLIM scores. P scores the raw target; D scores TRAIN-centred departures; CLIM is the source-free P/ROAD phase clock. Cov. is raw routing energy captured by roads. Errors pool held-out observations within each city.

City	Cov.	CLIM	P				D		
			UrbanFlow		Ridge	POD	UrbanFlow		Ridge
		ROAD	ROAD	FACE	FACE	FACE	ROAD	FACE	FACE
Porto	0.904	0.472	0.682	0.607	0.701	0.701	0.468	0.411	0.504
San Francisco	0.886	0.727	0.597	0.528	0.611	0.637	0.240	0.209	0.262
Chengdu	0.731	0.931	0.606	0.440	0.572	0.575	0.120	0.085	0.126

Note S6 Held-out coupling and baseline results

The three focal cities replicate a procedure, not a single operator transferred between cities. Porto has the strongest departure coupling. San Francisco retains positive D scores despite a raw climatology stronger than raw operator reconstruction. Chengdu separates a highly recurrent raw field from weak departure prediction: P/ROAD is 0.606, CLIM is 0.931, and D/ROAD is 0.120. Raw reproduction therefore cannot be read as source-specific information without the D comparison. The primary D entries have no reported paired interval or pairing-null result; Figure S2 shows their date-resolved values without supplying such an interval.

Full-face ridge and TRAIN-only POD equal or outperform the road restriction on FACE targets. D/FACE ridge scores are 0.504, 0.262 and 0.126, against road-restricted 0.411, 0.209 and 0.085. Roads therefore supply independently specified, interpretable support rather than the highest linear reconstruction accuracy. Interior-face counts in Porto, San Francisco and Chengdu are 1,984, 5,512 and 7,080; D-frame selected POD ranks are 958, 16 and 64. Table S2 consistently pools errors rather than averaging fold scores.

Three-city P-frame physical reconstruction. Figure S6 visualizes observed and predicted road-supported routing fields using the frozen P estimator with raw TEST inputs and targets. No phase climatology is added and no D departure is substituted. Porto and San Francisco preserve their archived `picked_i/picked_phase` snapshots. Chengdu selects the TEST date whose daily P error is closest to the median daily P error: `picked_i=2`, `picked_phase=6` (12:00 in the 06–23 h window), `picked_date=16304`. Its pinned-basis export is checked against the frozen R_{P}. These snapshots are qualitative complements to the complete pooled scores, not additional independent experiments.

Neural initialization and aggregation. Sealing initialization before network construction gives Graph WaveNet P/FACE means 0.548 in Porto (range 0.515–0.588), 0.221 in San Francisco (0.096–0.279) and 0.315 in Chengdu (0.260–0.350). Configuration choice is itself seed-dependent: Chengdu selects configurations $2, 0, 0, 0, 1$, Porto selects configuration 1 at four seeds and 0 at one, and San Francisco varies by fold and seed. Fifteen of the 100 San Francisco grid runs collapse to a zero field with VAL error exactly one; the minimum-VAL rule cannot select such a run when a non-collapsed alternative exists.

Define the Porto contrast as Graph WaveNet minus UrbanFlow. The paired-date value on the uncontrolled archived draw is -0.1102 (95% interval $[-0.1762, -0.0268]$, Holm-adjusted $p = 0.0088$). Under the sealed protocol the corresponding differences are -0.059 on the seed mean and -0.019

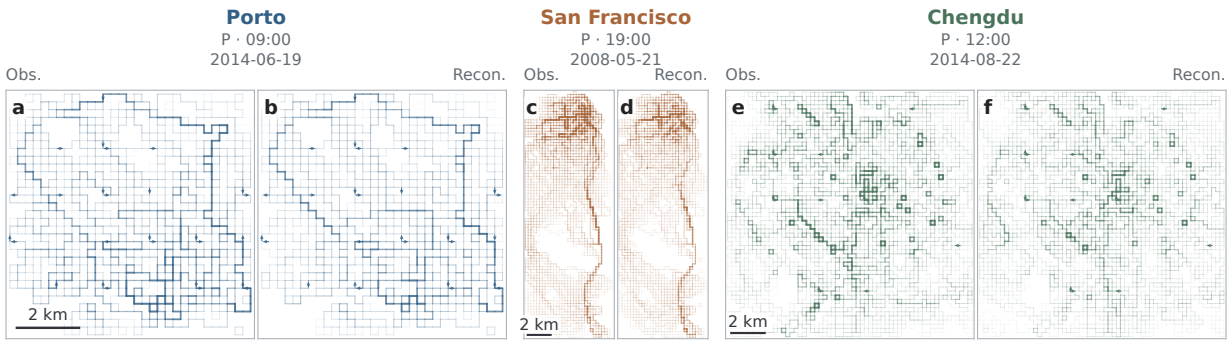


Figure S6: Three-city held-out physical reconstruction in the P frame. Observed and reconstructed road-supported routing fields for Porto, San Francisco and Chengdu, using the frozen P estimator with raw TEST inputs and targets; no phase climatology is added. Porto and San Francisco preserve their archived picked indices and phases. Chengdu uses the TEST date with daily P error closest to the median daily P error, specifically `picked_i=2`, `picked_phase=6` (12:00), `picked_date=16304`, and passes the pinned-basis/frozen-R_{P} export check. Observed and predicted panels within a city use identical glyph sites and the observed field's 99th-percentile magnitude scale, with saturation above it; colour scales are city-specific. The maps are qualitative, whereas Table S2 uses every held-out observation.

at the worst seed; the interval is not recomputed because sealed outputs retain pooled rather than per-date errors. UrbanFlow leads every realised seed in Porto and Chengdu by at least 0.019 and 0.090. San Francisco's sealed 0.221 is a fold-mean-of-R^2 summary, while UrbanFlow's 0.528 pools TEST errors; their difference is descriptive and is not used for a pooled like-for-like ranking. These results concern this completed adaptation and P/FACE protocol, not deep learning generally or a D-frame result.

Gravity and loading diagnostics. Native gravity CPC values are 0.525, 0.464 and 0.555. TEST-scaled gravity loading gives 0.136, 0.270 and 8.3×10^{-5}; exact-TEST-OD AON gives 0.172, 0.336 and 0.002, with fixed-amplitude values 0.126, −0.533 and −0.565. These tasks use different inputs and are not source-only accuracy rankings.

Chengdu's OD table contains 1.64 M TEST trips, of which 1.28 M remain after snapping. Its strong OD overlap but near-zero loading score is therefore not a ranking of demand quality. The loaded field has cosine 0.04 with the observed field, and destination permutation leaves the loaded field almost unchanged (cosine 0.93–0.94), showing that this pipeline resolves a largely margin-driven pattern. Rebuilding the observed three TEST dates from all fixes reproduces the frozen field at pooled cosine 0.999999. Using occupied runs alone removes 16.3% of absolute flux, leaves 76% of energy, and gives pooled cosine 0.929 with the all-fix field. Thus the activity-set difference is real and bounded, but by itself is unlikely to account for the loading mismatch. All loading rows diagnose the implemented demand–registration–loading pipeline rather than attainable performance of traffic assignment as a paradigm.

Note S7 Joint-spectrum, daily-score and coherent-response results

The joint field spectra, daily P/D scores and coherent response answer different questions. No operator is refitted for the spectral displays. The coherent comparison uses the same per-date DFT support as the joint spectrum in each city.

Three-city joint spectra. At the eight-mode display cut in Figure S1, source energy coverage is 0.6343, 0.9656 and 1.0000 in Porto, San Francisco and Chengdu, and routing coverage is 0.7989, 0.9892 and 1.0000. Chengdu's complete coverage follows from its eight-TRAIN-date snapshot-rank ceiling and does not demonstrate greater intrinsic low dimensionality. The common logarithmic floor is $10^{-4.5}$; all displayed cells exceed it. Chengdu's time-independent component carries 0.668 of source energy and 0.811 of full-face routing energy. The weighted spectra account for raw-block energy to machine precision; this is a numerical conservation check, not held-out evidence.

Held-out daily P/D scores. Figure S2 resolves the pooled entries in Table S2 by TEST date. Porto has 51 TEST dates, San Francisco has five blocked folds and Chengdu has three TEST dates in the continuous-observation protocol. The daily spread is descriptive: it does not replace the common pooled-error convention and does not supply the missing paired-date interval for the primary D comparison. In particular, Chengdu's three dates are insufficient for a strong date-heterogeneity claim.

Coherent TRAIN/TEST response. Figure S3 compares the frozen P-selected operator with observed coherent routing on both TRAIN and TEST. TRAIN panels describe the fit sample and are not independent validation. On TEST, the direction alignments γ_m at $m = 0$ are 0.978, 0.943 and 0.821 in Porto, San Francisco and Chengdu, respectively; at the first nonzero bin $m = 1$ they are 0.935, 0.908 and 0.628. The corresponding norm ratios ρ_m are 0.800, 0.612 and 0.709 at $m = 0$, and 0.790, 0.849 and 0.596 at $m = 1$. San Francisco values are fold means. Across all ten Chengdu bins, alignment spans 0.474–0.821 and norm ratio spans 0.596–0.747. These results show partial alignment in both direction and magnitude; they do not convert coherent-date averages into observation-level P/D scores.

Underpowered Chengdu full-FACE nonlinear exploration. Figure S7 reports an exploratory 18-phase A-to-D comparison on the complete FACE target. For each nonlinear arm, ΔR^2 is the mean-date full-FACE error of `linF` minus that of the arm, so positive values favour the nonlinear arm. The `quadF` contrast is -0.005854 with descriptive 95% interval $[-0.006105, -0.005661]$; the `krrF` contrast is $+0.000418$ with interval $[-0.000023, +0.000698]$, which crosses zero.

There are only three TEST dates, and D's all-phase linear design has 144 columns for 144 TRAIN rows and is therefore saturated.

The design is **underpowered**: these arms neither establish a nonlinear improvement nor support a claim of no nonlinearity.

Full-FACE scores must not be mixed with POD-internal scores. Although 9,999 paired-date bootstrap draws are recorded, three dates provide only 27 ordered date resamples, so the intervals are descriptive rather than confirmatory.

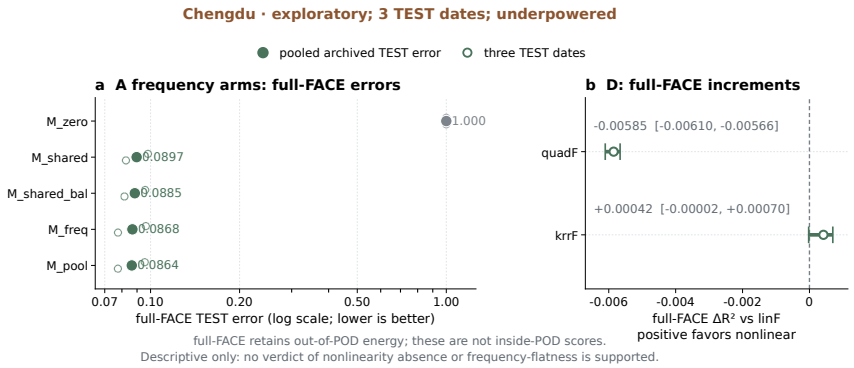


Figure S7: Underpowered exploratory Chengdu full-FACE nonlinear comparison. Full-FACE ΔR^2 relative to `linF` for `quadF` and `krrF` on the 18-phase A-to-D design, with descriptive 95% paired-date intervals. `quadF` gives -0.005854 $[-0.006105, -0.005661]$; `krrF` gives $+0.000418$ $[-0.000023, +0.000698]$. Only three TEST dates are available, and the 144-column linear design saturates the 144 TRAIN rows. The experiment is exploratory and underpowered; it does not demonstrate either the presence or absence of nonlinearity.

Note S8 Three-city anisotropy results

Median response anisotropy is 0.635, 0.651 and 0.705 in Porto, San Francisco and Chengdu, against structural-tensor medians 0.144, 0.057 and 0.280. The response exceeds structural anisotropy at 79.3% of Porto cells, 71–72% of San Francisco cells and 72.4% of Chengdu cells. These are properties of the fitted response on its identified source span, not of the instantaneous flow distribution. Figure S4 shows the full distributions behind these medians. Its zero-response atoms are distinct from low but nonzero anisotropy, and its rank-one atoms explain the vertical approach at $A = 1$.

Strong local anisotropy does not imply one city-wide preferred direction. Unweighted nematic coherence is 0.087 in Porto, 0.21–0.22 across San Francisco folds and 0.057 in Chengdu. Principal axes vary across space rather than collapsing onto one compass orientation.

Using the joint $A_C A_U$-weighted estimand in Eq. S11, structural-axis similarity is 0.79, 0.79 and 0.76 in Porto, San Francisco and Chengdu. San Francisco's anisotropy map is reproducible across five folds, with mean pairwise spatial correlation 0.984 (range 0.974–0.991) and anisotropy-weighted nematic axis agreement 0.993. The invariance checks concern scalar anisotropy and covariant axial/tensor transformations as specified in Note S4; they do not imply an axis that remains numerically unchanged under a rigid physical rotation.

Note S9 Road specificity and matched-capacity structural results

Rank-matched subspace specificity. The structural test uses each city's full aligned observation pool, not the held-out P/D pool. Full-pool road captures are 0.900, 0.900 and 0.731 in Porto, San Francisco and Chengdu. Every one of the 999 random conservation-compatible subspaces leaves more routing energy uncaptured than the real road subspace (0/999 at least as good, empirical $p = 1/1000$). The excess $\Delta E = E_{\text{null}} - E_{\text{road}}$ ranges from 0.23 to 0.30 in Porto, 0.28 to 0.33 in San Francisco and 0.09 to 0.15 in Chengdu. This is a null distribution, not a confidence interval. The ambient $\ker D$ and rank are fixed while the random subspaces vary; the result concerns representation specificity and does not show that a source predicts the state within the road space.

Matched-capacity media. At $R = n_{\text{int}} = (1984, 5512, 7080)$ nominal pre-normalization block values, the constant-medium held-out R^2_{cond} values are 0.613, 0.743 and −0.030. Road-medium gains over that reference are 0.009, 0.010 and −0.001; mobility-inferred gains are 0.092, 0.085 and 0.036; and permuted-mobility-medium gains are −0.047, −0.052 and −0.016. The paired mobility-minus-road differences are 0.083 in Porto (95% interval 0.060–0.109), 0.075 in San Francisco (0.062–0.091) and 0.038 in Chengdu (0.035–0.040). Figure S5 shows all three media at this capacity.

The comparison matches capacity and estimator, not information: the road pattern is an external GIS raster, whereas the mobility-inferred pattern is estimated from TRAIN mobility. Both use the same nominal block capacity under the same normalization. Permuting the mobility medium's spatial arrangement preserves its values, partition and nominal block count and removes the gain. Chengdu uses one all-motion channel, a frozen per-fold basis and four held-out alternate-hour dates; its interval resamples only those dates. The design cannot separate the influence of date count, delivery phase and channel, and the four-date interval is only a descriptive summary, not the 14-date P/D experiment.

Together, the two tests show that the road-induced subspace is not generic under the stated rank-matched null, while the road-only medium does not match the mobility-inferred medium under this estimator and information regime. They do not show that road geometry lacks physical meaning, that all alternative road structures have been tested, or that source-only closure has been achieved. The source files, real R grids and missing-module definition gap are listed in Notes S4 and S10.

Supplementary Reproducibility

Note S10 Artifact map and reproducibility

Table S3 separates frozen-score recomputation, figure rebuilding and incomplete raw/training exports. All paths are repository-relative. A file's presence does not imply that an end-to-end raw-trajectory rebuild is possible, and missing modules or per-run outputs are stated rather than replaced by inferred definitions.

Deep-baseline record. The archived scoring chain is `run_modern.py` → `bmlib.anomalies` → `cc_lib.anomaly`. Only TRAIN rows are phase-centred by the called implementation, leaving VAL/TEST raw. Deduplicating Porto GWNET fold 0 by retaining the final authoritative per-date record gives an archived single draw of 0.4970176832 in Porto; this is not the sealed seed mean. Graphormer's 60-epoch budget and incomplete fold-level prediction-file export remain disclosed. Per-date error records exist for all scored archived folds, but zero-divergence residuals follow from projection and are not learned conservation.

Release scope and unresolved checks. The public export is intended to separate figure rebuilding from frozen products, core-score recomputation and end-to-end raw-trajectory reconstruction. It contains no individual-level trajectory records. Large prediction files, unavailable original training artifacts, the missing matched-capacity modules named above, and incomplete sealed per-run outputs must not be represented as included. Raw-data redistribution requirements, OSM snapshot/version and per-artifact licence notices remain release checks (Note S1). Public repository location: https://github.com/Wangwangw233/paper-urbanflow.

Table S3: Artifact and reproducibility map. Paths are relative to the repository root. The status column distinguishes archived evidence from entry points runnable in the present export.

Result family	Source artifacts and code	Reproduction status or limitation
P/D/CLIM and linear controls	`reports/mainstream_mobility_comparison/crosscity_results.csv`; `reports/mainstream_mobility_comparison/results/s1/*.json`; `reports/mainstream_mobility_comparison/scripts/s1_crosscity.py`	Frozen per-fold objects and pooled scores; no primary-D interval or pairing-null result is supplied.
Daily scores and coherent response	`reports/mainstream_mobility_comparison/results/s1/*.json`; `reports/spatiotemporal_constitutive_spectra/results/obj_*.json`; `reports/threecity_redesign/results/chengdu_response18.npz`	Frozen date-level P/D diagnostics and coherent objects; Chengdu uses all ten 18-point DFT bins, and San Francisco fold aggregation remains distinct from pooled scoring. The present Chengdu response file contains TEST only; a TRAIN display requires a same-protocol TRAIN export rather than inference from TEST.
Three-city physical reconstruction	`crosscity_sample_complexity/artifacts/routing_snapshot_Porto.npz`; `crosscity_sample_complexity/artifacts/routing_snapshot_SF.npz`; `crosscity_sample_complexity/artifacts/routing_snapshot_meta.json`; `reports/threecity_redesign/results/chengdu_snapshot18.npz`; `reports/threecity_redesign/results/chengdu_frozen_response_manifest.json`; `reports/threecity_redesign/scripts/chengdu_frozen_response.py`	All three snapshots use the P estimator with raw TEST inputs and targets. Chengdu passes pinned-basis and frozen-R_{P} checks and archives `picked_i=2`, `picked_phase=6`, `picked_date=16304`. Qualitative snapshots do not replace complete held-out scores.
Chengdu nonlinear exploration	`reports/chengdu_nonlinear_exploratory/config.json`; `reports/chengdu_nonlinear_exploratory/scripts/run_experiment.py`; `reports/chengdu_nonlinear_exploratory/results/chengdu_nonlinear_exploratory.json`; `reports/chengdu_nonlinear_exploratory/results/validation.json`	Sixteen validation checks pass. The 18-phase full-FACE A-to-D experiment is underpowered: three TEST dates, 144 columns/144 TRAIN rows, and only 27 ordered date resamples. It supports neither nonlinear improvement nor absence of nonlinearity; full-FACE and POD-internal scores remain distinct.
Archived neural draw	`reports/deep_baseline_training_record/MANIFEST.csv`; `reports/deep_baseline_training_record/artifacts/_perdate.jsonl`; `reports/deep_baseline_training_record/artifacts/modern_results_all.csv`; `reports/deep_baseline_training_record/scripts/run_modern.py`	Twenty copied files were MD5-checked with no mismatch. Large predictions, Jacobians and logs are not included; scripts require legacy-root configuration.
Sealed Graph WaveNet	`reports/table_fill_chengdu/README.md`; `reports/table_fill_chengdu/scripts/gwnet_fixed.py`; `reports/table_fill_chengdu/scripts/seal_proof.py`	Seed means, ranges and initialization proof are summarized; complete sealed per-run JSON outputs are not present in this export. San Francisco remains fold-mean aggregation.
Gravity and AON	`reports/mainstream_mobility_comparison/results/s3/`; `reports/mainstream_mobility_comparison/results/s4/`; `reports/table_fill_chengdu/README.md`	Archived JSON artifacts and the Chengdu collated record support the three-city scores; loading tasks remain distinct from source-only inference.
Joint three-city spectrum	`reports/threecity_joint_spectrum/manuscript_revision/verified_joint.npz`; `reports/threecity_joint_spectrum/manuscript_revision/audit.json`; `reports/threecity_joint_spectrum/manuscript_revision/redraw_joint_spectrum.py`	Weighted Parseval display, source hashes and render checks are archived. Figure rebuilding does not refit an operator.
Anisotropy	`full_operator_spatial_anisotropy/results/fo4_fig5_data.npz`; `results/kr_response_characteristics/`; `reports/threecity_figure5_revision/results/chengdu_anisotropy_summary.json`	Frozen arrays and Chengdu D/P sensitivity records support the three-city maps and distributions.
Structural null	`final_ablation/results/partA_topology.json`; `reports/threecity_figure5_revision/results/partA_chengdu.json`; `reports/threecity_figure5_revision/scripts/cd_partA_null.py`	999 draws per city; Chengdu includes a Porto control rerun to machine precision. Full-pool representation test only.
Matched-capacity media	`benchmark/scripts/bench_lib.py`; `benchmark/domain_rankmatched_results.csv`; `benchmark/domain_rankmatched_pairwise.csv`; `reports/threecity_figure5_revision/results/partF_summary.json`; `reports/threecity_figure5_revision/scripts/cd_partF_sweep.py`	The passing-cell record is not a current-export run. `domain_rankmatched.py` and `domain_lib.py` are absent, the wrapper retains a legacy absolute root, and required aligned fields, date-freeze files and GIS rasters are unavailable. Only frozen results and figure rebuilding are supported.
Canonical supplementary figures	`paper/merged_manuscript/figures_supp/README.md`; `reports/threecity_figure5_revision/scripts/fig5_redesign.py`; manuscript figure scripts	Seven canonical PDFs map one-to-one to Figures S1–S7; figure rebuilding is distinct from recomputing the experiments.

References

[1] Luis Moreira-Matias, João Gama, Michel Ferreira, João Mendes-Moreira, and Luís Damas. Taxi service trajectory: Prediction challenge, ECML PKDD 2015. UCI Machine Learning Repository, dataset 339, 2015.

[2] Michal Piorkowski, Nataša Sarafijanovic-Djukic, and Matthias Grossglauser. CRAWDAD dataset epfl/mobility (v. 2009-02-24). CRAWDAD archive, 2009.

[3] Didi Chuxing. GAIA open dataset. https://gaia.didichuxing.com/, 2014. Chengdu ride-hailing trajectory sample, August 2014; exact release version not recorded.

[4] Mordechai Haklay and Patrick Weber. OpenStreetMap: user-generated street maps. *IEEE Pervasive Computing*, 7(4):12–18, 2008.

[5] Zonghan Wu, Shirui Pan, Guodong Long, Jing Jiang, and Chengqi Zhang. Graph WaveNet for deep spatial-temporal graph modeling. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence*, pages 1907–1913, 2019.

[6] Filippo Simini, Marta C. González, Amos Maritan, and Albert-László Barabási. A universal model for mobility and migration patterns. *Nature*, 484(7392):96–100, 2012.

[7] Filippo Simini, Gianni Barlacchi, Massimilano Luca, and Luca Pappalardo. A Deep Gravity model for mobility flows generation. *Nature Communications*, 12:6576, 2021.

[8] John Glen Wardrop. Some theoretical aspects of road traffic research. *Proceedings of the Institution of Civil Engineers*, 1(3):325–362, 1952.